

NEW BRAUNFELS, TX, USA

WMO: 722416

Lat: 29.709N

Lon: 98.046W

Elev: 645

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 12971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.7	31.8	12.2	10.5	41.2	16.4	12.9	45.7	26.7	54.3	24.5	53.6	8.3	20	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.3	100.3	74.1	98.6	74.2	96.7	74.4	78.3	89.5	77.6	88.5	77.1	87.8	9.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	138.8	80.5	75.2	135.6	80.2	74.6	132.7	80.0	42.1	90.0	41.4	88.8	40.8	87.6	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.5	18.6	DB	22.4	103.8	4.0	3.3	19.5	106.2	17.1	108.2	14.9	110.0	12.0	112.5	
			WB	19.8	79.5	3.6	1.0	17.2	80.2	15.2	80.8	13.1	81.3	10.6	82.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	51.9	56.0	62.9	69.9	76.4	82.3	83.9	85.2	80.0	71.0	61.1	53.2	(6)
(7)		DBStd	13.74	9.14	10.08	8.59	6.88	5.58	3.44	3.05	3.08	5.08	7.72	9.10	8.98	(7)
(8)		HDD50	233	86	50	11	1	0	0	0	0	0	1	16	68	(8)
(9)		HDD65	1464	416	278	144	31	3	0	0	0	1	35	178	378	(9)
(10)		CDD50	7370	144	219	409	598	818	969	1051	1093	899	652	349	169	(10)
(11)		CDD65	3125	9	27	77	178	356	519	586	628	450	221	60	14	(11)
(12)		CDH74	33583	71	197	544	1458	3320	5832	6911	7924	4730	2059	451	86	(12)
(13)		CDH80	16487	6	37	123	459	1375	2962	3769	4544	2364	757	84	7	(13)
(14)	Wind	WSAvg	8.7	9.1	9.8	10.1	10.3	9.5	8.4	7.7	7.7	7.2	7.9	8.4	8.6	(14)
(15)	Precipitation	PrecAvg	35.0	2.0	2.2	2.0	2.6	4.7	4.3	1.8	2.7	3.6	4.3	2.9	2.4	(15)
(16)		PrecMax	58.4	9.7	9.1	5.2	9.1	15.8	14.1	5.8	10.5	10.5	26.8	10.8	17.5	(16)
(17)		PrecMin	18.5	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.3	0.0	0.0	0.0	(17)
(18)		PrecStd	10.8	1.8	1.7	1.1	1.9	3.6	3.5	1.7	2.4	2.4	4.3	2.4	3.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.4	85.0	87.3	92.9	96.5	101.3	101.8	102.8	101.0	93.2	86.1	80.8	(19)
(20)		MCWB	61.9	63.9	66.3	68.3	74.1	73.8	73.6	74.6	72.8	71.3	70.5	65.6	(20)	
(21)		2%	DB	75.5	79.6	83.4	88.4	93.2	97.7	99.4	100.5	97.3	90.1	82.6	76.2	(21)
(22)		MCWB	61.4	64.2	65.3	69.6	73.3	74.1	74.2	74.4	73.5	72.2	67.5	63.5	(22)	
(23)		5%	DB	71.9	75.6	80.1	84.8	90.3	95.1	97.4	99.0	94.4	87.6	79.4	72.8	(23)
(24)		MCWB	60.3	62.0	65.6	68.7	73.5	74.4	74.6	74.5	73.3	70.7	66.0	62.9	(24)	
(25)		10%	DB	68.2	71.9	76.5	82.0	87.8	92.6	95.1	97.1	91.5	84.5	75.9	69.5	(25)
(26)		MCWB	59.3	61.9	64.1	68.0	72.8	74.7	74.8	74.6	73.0	69.5	65.5	61.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	69.5	71.6	75.5	78.3	79.3	78.8	78.5	78.4	77.4	74.0	70.1	(27)
(28)		MCDWB	72.8	76.1	79.0	85.1	89.0	90.8	91.0	91.3	87.7	84.1	80.3	75.4	(28)	
(29)		2%	WB	66.5	68.0	69.8	73.4	77.0	78.2	78.0	77.8	77.4	75.9	71.8	68.3	(29)
(30)		MCDWB	71.3	73.9	76.9	82.0	87.0	89.1	89.6	90.0	86.6	82.8	77.5	73.0	(30)	
(31)		5%	WB	64.1	66.5	68.5	72.2	75.8	77.4	77.3	77.2	76.6	74.8	70.0	66.4	(31)
(32)		MCDWB	69.2	71.9	75.2	80.1	85.1	87.8	88.5	89.0	85.8	81.7	75.9	70.8	(32)	
(33)		10%	WB	61.0	64.7	67.3	71.0	74.8	76.6	76.6	76.6	75.8	73.4	68.1	63.4	(33)
(34)		MCDWB	66.2	69.6	73.5	78.2	83.5	86.5	87.3	88.2	85.0	80.3	74.3	67.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	22.3	22.1	22.2	21.8	20.8	20.8	21.9	23.3	22.0	22.9	22.6	22.3	(35)	
(36)		5% DB	MCDBR	27.7	26.3	24.5	24.5	23.2	24.0	25.4	25.9	25.0	23.6	24.2	26.6	(36)
(37)		MCWBR	16.3	14.9	12.1	10.6	8.1	5.9	5.6	5.6	7.3	9.5	12.6	16.7	(37)	
(38)		5% WB	MCDBR	20.8	19.6	19.5	18.2	18.6	19.1	20.6	22.3	20.0	18.5	18.3	21.0	(38)
(39)		MCWBR	14.7	13.5	11.4	9.3	7.7	6.1	5.5	5.4	6.5	9.0	11.0	16.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.335	0.362	0.401	0.449	0.430	0.441	0.427	0.416	0.375	0.363	0.319	(40)	
(41)		taud	2.471	2.482	2.418	2.325	2.238	2.343	2.344	2.368	2.351	2.452	2.420	2.533	(41)	
(42)		Ebn at Noon	285	296	293	284	269	273	269	273	272	278	272	286	(42)	
(43)		Edh at Noon	30	32	36	41	45	40	40	39	38	32	31	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	954	1111	1419	1695	1832	2087	2101	1998	1645	1367	1027	845	(44)	
(45)		RadStd	105	199	136	160	169	133	178	111	160	180	116	108	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.86	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +344	(m) +263	(46)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	(47)

Nomenclature: See separate page